

# A Finite Volume Multilevel WENO Scheme for Multidimensional Problems

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Received: date / Accepted: date

**Abstract** ???

**Keywords** ??? · weighted essentially non-oscillatory · ML-WENO · hyperbolic equation

**Mathematics Subject Classification (2010)** ??? · 65D12 · 65M12 · 76M12

## 1 Introduction

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Challenges. – Identify good stencils automatically. If not, — add elements to the stencil? (Not easy!) — reduce polynomial order. – Current techniques are two-level, requiring — a single large stencil — many small stencils of the same polynomial degree – Want to allow constant polynomials (one element stencils)

We define a new multi-level reconstruction – Can use any combination of stencils – Identifies good stencils – Combines polynomials of any order – Picks out the highest order of accuracy of stencils without a shock

Finite volume: Minimal number of DOFs per mesh element.

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The first author was funded in part by the U.S. National Science Foundation grant DMS-1912735. The second author was funded by the Taiwan Ministry of Science and Technology grant MOST 109-2115-M-110-003-MY3 and the National Center for Theoretical Sciences, Taiwan.

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1. New ML-WENO
2. Selection of good stencils
3. New polynomial smoothness indicator

The condition number of the approximation problem cannot be too large!

In this paper, we develop novel finite volume, multilevel weighted essentially non-oscillatory reconstructions with adaptive order (ML-WENO-AO) suitable for use in multiple space dimensions. Our multilevel reconstructions seamlessly weight the stencil polynomial approximations of various degrees to bias away from oscillatory polynomials and smooth polynomials of low degree, thereby selecting the smooth polynomial(s) of maximal degree of approximation. We also apply these new reconstructions to develop a scheme for solving a system of hyperbolic conservation laws,

$$u_t + \nabla \cdot F(u) = 0, \quad t > 0, \mathbf{x} \in \mathbb{R}^d, \quad (1.1)$$

where  $u = u(\mathbf{x}, t)$  and  $d > 0$  is an integer.

WENO schemes [7, 11, 9] are one of the more popular and effective set of numerical methods for solving hyperbolic conservation laws.

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- (1) you can take a bad stencil and simply drop the polynomial degree
  - (2) handle BCs (need constants!)
  - (3) include constants (if there is simply no good stencil!)
  - (4) Simplicity of implementation for unstructured meshes
- Similar work on WENO-AO [3, 2]

The paper is organized as follows. ??? Finally, a summary and conclusions are provided in Section 7.

## 2 Selection of the Stencil Polynomials

Defining polynomials on stencils in 2 and 3 dimensions has been seen to be problematic [2]. However, the process can be automated as we describe in this section. But first, we make precise some of the terms and notation we use in the multidimensional setting.

Let  $d$  be the dimension of the space variables. An *element*  $E \subset \mathbb{R}^d$  is a bounded, closed subset that is homeomorphic to a closed ball (i.e., the closure of its interior is the element and the interior is simply connected), such as a polytope. Let the domain  $\Omega \subset \mathbb{R}^d$  and  $h > 0$  be given. Let  $\mathcal{T}_h$  be a quasiuniform mesh of elements that tessellate  $\Omega$  such that  $h_E = \text{diam}(E) \leq h$  for each element  $E \in \mathcal{T}_h$ .

By saying that the elements tessellate the domain, we mean that their union is the closure of  $\Omega$  and they intersect only on their boundaries. Recall that as  $h \rightarrow 0^+$ , we write

$$f(h) = \Theta(h) \iff c_1 h \leq |f(h)| \leq c_2 h, \quad (2.1)$$

for positive constants  $c_1$  and  $c_2$  when  $h > 0$  is sufficiently small. Quasiuniformity of the mesh means that  $h_E = \Theta(h)$  uniformly as  $h \rightarrow 0^+$ .

A mesh *stencil*  $S = \{E_0, E_1, \dots, E_L\}$  is a subset of elements of  $\mathcal{T}_h$  such that their union  $\Omega_S = \cup_{k=0}^L E_k$  is a bounded, closed subset that is homeomorphic to a closed ball.

We are given a function  $u : \Omega \rightarrow \mathbb{R}$  called the *solution*, since it is usually the solution to a differential equation. Its element averages are

$$\bar{u}_E = \frac{1}{|E|} \int_E u(\mathbf{x}) d\mathbf{x} \quad \forall E \in \mathcal{T}_h. \quad (2.2)$$

We denote the space of polynomials of degree  $r \geq 0$  in  $d$  variables by  $\mathbb{P}_r$ .

## 2.1 Stencil polynomials

Given a stencil  $S$  of the mesh and an index  $r \geq 1$ , our goal is to define an associated *stencil polynomial*  $P \in \mathbb{P}_{r-1}$ . To increase the numerical stability of the computations, we represent  $P$  as

$$P(\mathbf{x}) = \sum_{|\alpha| < r} c_\alpha \left( \frac{\mathbf{x} - \mathbf{x}_S}{h_S} \right)^\alpha, \quad (2.3)$$

where  $\mathbf{x}_S$  is the center of  $\Omega_S = \bigcup_{E \in S} E \subset \mathbb{R}^d$ ,  $h_S = \max\{h_E : E \in S\}$ , and  $\alpha$  is a  $d$ -dimensional multi-index.

In the finite volume context, we require that the “mass” of  $u$  be preserved for each  $E \in S$ ,

$$\frac{1}{|E|} \int_E P(\mathbf{x}) d\mathbf{x} = \bar{u}_E, \quad (2.4)$$

which leads directly to the linear system

$$A\mathbf{c} = \mathbf{u} \quad (2.5)$$

for the vector of polynomial coefficients  $\mathbf{c} = (c_\alpha)$  given the vector of solution averages  $\mathbf{u} = (\bar{u}_E)$ .

The matrix  $A$  is  $L \times N$  where  $N = \binom{r+1}{2}$  is the number of coefficients and  $L \geq N$  is the number of elements in the stencil. The linear system is generally over-determined, i.e.,  $L > N$ , so we need to find the least squares solution by solving the normal equations  $A^T A \mathbf{c} = A^T \mathbf{u}$ . We have just defined a linear projection operator  $\pi_S$  taking  $u$  to  $\pi_S u = P$  as given in (2.3).

However, it is not enough to merely define the polynomial  $P$ . We also require that it approximate the solution accurately. To be precise, each stencil under consideration  $S = S^{(r)}$  must have an index  $r \geq 1$  such that its associated stencil polynomial  $P \in \mathbb{P}_{r-1}$  satisfies

$$|\mathcal{D}^\alpha u(\mathbf{x}) - \mathcal{D}^\alpha P(\mathbf{x})| \leq \begin{cases} Ch^{r-|\alpha|} & \text{if } u \text{ is smooth on the stencil,} \\ C & \text{otherwise,} \end{cases} \quad \forall \mathbf{x} \in E \in S^{(r)}, \quad (2.6)$$

for some constant  $C > 0$  independent of the mesh as  $h \rightarrow 0^+$ , where  $\mathcal{D}^\alpha$  is partial differentiation of order  $\alpha$ .

The Bramble-Hilbert Lemma [4,6] tells us that  $P$  will approximate  $u$  and its derivatives provided that two conditions hold. First,  $\pi_S$  must preserve polynomials,

which it clearly does. Second,  $\pi_S$  must be a bounded operator, which it will be as long as the matrix  $A^T A$  is well conditioned and the mesh is quasiuniform.

We propose finding the compact singular value decomposition (SVD) and the singular values  $s_1 \geq s_2 \geq \dots \geq s_M > 0$  of  $A$ , i.e.,

$$A = U \Sigma V^T, \quad \Sigma = \begin{bmatrix} s_1 & 0 & \dots & 0 \\ 0 & s_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & s_M \end{bmatrix}. \quad (2.7)$$

The condition number of the least squares matrix  $A^T A$  is infinite if  $M < N$  (i.e., if the system is singular and cannot be solved) and  $(s_1/s_N)^2$  otherwise (i.e.,  $M = N$ ). In double precision floating point arithmetic, to maintain accuracy we would want perhaps  $(s_1/s_N)^2 < 1e+8$ . If this condition fails, we should reject the approximation  $P$ .

In the case of a rejected  $P$ , one could add elements to the stencil, but it is unclear how one might do this reliably and automatically in more than one space dimension. Instead, we propose to reduce the order of the approximation ( $r$  is replaced by  $r-1$ ) and try again. Note that if we drop order to  $r=1$  (so  $P \in \mathbb{P}_0$  is a constant), we will have a  $1 \times 1$  matrix  $A^T A$  with a good condition number, and so our algorithm will terminate.

In practice, one generally evaluates the stencil polynomial on some *target element*  $E_0 \in S$ . In this case, it is important to add a constraint to the least squares solution. We require that  $P$  satisfy the “mass” constraint exactly on  $E_0 \in S$ . We can write this as

$$\frac{1}{|E_0|} \int_{E_0} P(\mathbf{x}) d\mathbf{x} = \bar{u}_{E_0} \iff \beta \cdot \mathbf{c} = \bar{u}_{E_0}, \quad (2.8)$$

where  $\beta^T$  is the row of  $A$  corresponding to element  $E_0$ . Let  $\mathbf{e}_0$  denote the standard unit vector that is one for element  $E_0$  and zero otherwise. Then the constraint is

$$\beta \cdot \mathbf{c} = \bar{u}_{E_0} = \mathbf{e}_0 \cdot \mathbf{u}, \quad \beta = A^T \mathbf{e}_0. \quad (2.9)$$

The constrained problem is

$$\begin{bmatrix} A^T A & \beta \\ \beta^T & 0 \end{bmatrix} \begin{pmatrix} \mathbf{c} \\ \lambda \end{pmatrix} = \begin{pmatrix} A^T \mathbf{u} \\ \bar{u}_{E_0} \end{pmatrix}. \quad (2.10)$$

This is easily solved in terms of the unconstrained least squares solution of (2.5),

$$\hat{\mathbf{c}} = B\mathbf{u}, \quad B = (A^T A)^{-1} A^T = V \Sigma^{-2} V^T A^T = V \Sigma^{-1} U^T, \quad (2.11)$$

and the vector

$$\alpha = (A^T A)^{-1} \beta = B\mathbf{e}_0. \quad (2.12)$$

The result is

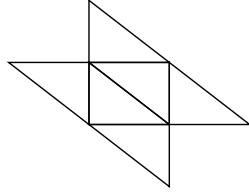
$$\mathbf{c} = \hat{\mathbf{c}} + \frac{1}{\beta \cdot \alpha} (\bar{u}_{E_0} - \beta \cdot \hat{\mathbf{c}}) \alpha. \quad (2.13)$$

We recap the procedure in the context of a time-stepping loop. We start with a stencil  $S$  and desired polynomial degree  $r-1$ . Before entering the time-stepping loop,

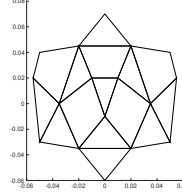
the matrix  $A$  is precomputed along with its compact SVD,  $A = U\Sigma V^T$ . If the normal matrix  $A^T A$  is singular or badly conditioned, we reject the polynomial, replace  $r$  by  $r - 1$ , and try again with a new matrix  $A$ . The process will terminate with  $r \geq 1$ , which is the index representing the highest possible polynomial degree that provides accurate numerical approximation on the stencil. The matrix  $B = V\Sigma^{-2}V^T A^T = V\Sigma^{-1}U^T$  can be precomputed as well. Within the time stepping loop, we use the precomputed  $A$  and  $B$  to find the vectors  $\hat{\mathbf{c}} = B\mathbf{u}$ ,  $\beta = A^T \mathbf{e}_0$ , and  $\alpha = B\mathbf{e}_0$ . We then quickly solve (2.13) for the vector of coefficients of the polynomial  $P$  for the given target element  $E_0$  corresponding to  $\mathbf{e}_0$ .

## 2.2 Examples of Bad Stencils

It is not difficult to find poor stencils for a given value of  $r$ . We depict in Figure 2.1 a stencil of six elements that leads to a singular matrix  $A$  when  $r = 3$  (note that  $\dim \mathbb{P}_2 = 6$ ). In Figure 2.2, we depict a stencil of 15 elements that has a large condition number when  $r = 5$  (note that  $\dim \mathbb{P}_4 = 15$ ). The extreme singular values are in fact  $s_1 = 4.174$  and  $s_{15} = 3.391\text{e-}17$ , so the condition number  $(s_1/s_{15})^2$  is  $1.515\text{e+}34$ . From the figures it is not obvious to the authors that either stencil should perform so poorly.



**Fig. 2.1** A stencil of 6 elements that leads to a singular matrix when  $r = 3$ .



**Fig. 2.2** A stencil of 15 elements that leads to a badly conditioned matrix when  $r = 5$ .

When we drop the index of the second stencil to  $r = 4$ , the extreme singular values are  $s_1 = 4.106$  and  $s_{10} = 0.114$ , so the condition number is only  $(s_1/s_{10})^2 = 1303.97$ . We report in Table 2.1 a study of the accuracy and convergence rate of the approximations using  $r = 5$  and  $r = 4$ . The diameter of the initial stencil is about 0.14, and each refinement level shrinks the mesh by a factor of 2. The average  $L^1$ -error is computed on the target element, which is the center one. We see that the 4th degree polynomial is unacceptably inaccurate, while the 3rd order polynomial shows good accuracy and 4th order convergence.

Consider now  $r = 3$  and the stencil of 6 square elements given by the meshpoints  $(0.1i, 0.1j)$  for  $i = 0, 1, 2, 3$  and  $j = 0, 1, 2$ . Clearly the matrix  $A^T A$  is singular. However, if we perturb the point  $(0.2, 0.1)$  to  $(0.2, 0.1 + \theta)$  for small  $\theta > 0$ , the problem is no longer singular. To the naked eye, however, there is little difference in the stencils for  $\theta = 1\text{e-}2$ ,  $1\text{e-}4$ , and  $1\text{e-}8$ , so we would not expect any of these stencils to perform acceptably.

**Table 2.1** Approximation of  $2(1 + \cos(2\pi x)) \exp(xy - y)$  on refined stencils congruent to the one depicted in Figure 2.2 using  $r = 5$  and  $r = 4$ . The target element is the center one, and the average  $L^1$ -error is computed on this element.

| refinement level | $r = 5, \mathbb{P}_4$ |      | $r = 4, \mathbb{P}_3$ |      |
|------------------|-----------------------|------|-----------------------|------|
|                  | $L^1$ -error          | rate | $L^1$ -error          | rate |
| 0                | 4.124e+07             | —    | 5.697e-06             | —    |
| 1                | 1.291e+06             | 5.00 | 3.550e-07             | 4.00 |
| 2                | 4.036e+04             | 5.00 | 2.218e-08             | 4.00 |
| 3                | 1.261e+03             | 5.00 | 1.386e-09             | 4.00 |
| 4                | 3.947e+01             | 5.00 | 8.658e-11             | 4.00 |

**Table 2.2** Approximation of  $2(1 + \cos(2\pi x)) \exp(xy - y)$  for  $r = 3$  on refined stencils given by the initial meshpoints  $(0.1i, 0.1j)$  for  $i = 0, 1, 2, 3$  and  $j = 0, 1, 2$ ,  $(i, j) \neq (2, 1)$  and  $(0.2, 0.1 + \theta)$ . The condition number  $\kappa$  is given, as well as the average  $L^1$ -error computed on the lower middle (target) element centered at  $(0.15, 0.05)$ .

| refinement level | $\theta = 1e-2$<br>$\kappa = 3.18312e+07$ |      | $\theta = 1e-4$<br>$\kappa = 3.13172e+11$ |      | $\theta = 1e-8$<br>$\kappa = 3.13121e+19$ |      |
|------------------|-------------------------------------------|------|-------------------------------------------|------|-------------------------------------------|------|
|                  | $L^1$ -error                              | rate | $L^1$ -error                              | rate | $L^1$ -error                              | rate |
| 0                | 1.455e-00                                 | —    | 1.452e+02                                 | —    | 1.452e+06                                 | —    |
| 1                | 4.803e-01                                 | 1.60 | 4.799e+01                                 | 1.60 | 4.799e+05                                 | 1.60 |
| 2                | 7.745e-02                                 | 2.63 | 7.740e+00                                 | 2.63 | 7.743e+04                                 | 2.63 |
| 3                | 1.064e-02                                 | 2.86 | 1.064e+00                                 | 2.86 | 1.065e+04                                 | 2.86 |
| 4                | 1.385e-03                                 | 2.94 | 1.384e-01                                 | 2.94 | 1.378e+03                                 | 2.95 |

We show in Table 2.2 a study of the condition number and approximation rates of these stencils for the aforementioned  $\theta$ . We see third order convergence for all stencils, but only the stencil using  $\theta = 0.01$  could be considered to give an accurate stencil polynomial. This stencil is also the only one with condition number meeting our requirement  $\kappa < 1e+8$ .

### 3 Efficient Stencil Polynomial Smoothness Indicators

We measure the smoothness of  $u(\mathbf{x})$  on the stencil  $S^{(r)} = \{E_0, E_1, \dots, E_L\}$  with respect to the target element  $E_0$  by a *smoothness indicator*  $\sigma$ . A smoother  $u$  registers a smaller  $\sigma$ , so it in fact measures *roughness*. One requires that as  $h \rightarrow 0^+$ , there is some  $D \geq 0$  such that

$$\sigma = \begin{cases} Dh^2 + \mathcal{O}(h^3) & \text{if } u \text{ is smooth on the stencil,} \\ \Theta(1) & \text{if } u \text{ has a jump discontinuity on the stencil.} \end{cases} \quad (3.1)$$

Unfortunately, any reasonable definition of  $\sigma$  would give a value that is not necessarily bounded below when there is a jump discontinuity in  $u$ , i.e., we have merely  $\sigma = \mathcal{O}(1)$  in this case. Fortunately, this issue does not cause problems when solving conservation laws, since it may occur only sporadically in time. To obtain (3.1), we would need to make an assumption, for example, that the discontinuity is bounded away from the boundaries of the elements [1]. However, it is simpler and common practice to justify (3.1) in the case that  $u$  is smooth and to make the assumption that the other case holds.

### 3.1 The classic smoothness indicator $\sigma_{JS}$

The classic way to define  $\sigma$  is due to Jiang and Shu [8], which can be extended to  $d$  space dimensions as

$$\sigma_{JS} = \sum_{1 \leq |\alpha| \leq r-1} \frac{h_0^{2|\alpha|}}{|E_0|} \int_{E_0} |\mathcal{D}^\alpha P(\mathbf{x})|^2 d\mathbf{x}, \quad (3.2)$$

where  $h_0 = h_{E_0} = \Theta(h)$ . If  $r = 1$  so  $P \in \mathbb{P}_0$  (e.g.,  $S^{(r)}$  has only one element), then the formula implies that  $\sigma_{JS} = 0$ .

#### 3.1.1 An estimate of $\sigma_{JS}$ in the smooth case

When  $u$  is smooth on the stencil, the estimate  $\sigma_{JS} = Dh^2 + \mathcal{O}(h^3)$  is known to hold in one space dimension [8], where  $D \geq 0$  is related to the derivative of  $u$ . The following lemma gives a similar estimate in multiple dimensions, and so provides the property required in (3.1).

**Lemma 3.1** *Suppose  $u$  is approximated on the stencil  $S^{(r)}$  by the polynomial  $P \in \mathbb{P}_{r-1}$ . If  $r \geq 2$ , then as  $h \rightarrow 0^+$ ,*

$$\sigma_{JS} = Dh_0^2 + \mathcal{O}(h^3), \quad \text{if } u \text{ is smooth on the stencil,} \quad (3.3)$$

where

$$D = \frac{1}{|E_0|} \int_{E_0} |\nabla u(\mathbf{x})|^2 d\mathbf{x} \geq 0. \quad (3.4)$$

*Proof* Starting from (3.2), note that

$$\sigma_{JS} = \frac{h_0^2}{|E_0|} \int_{E_0} |\nabla P(\mathbf{x})|^2 d\mathbf{x} + \sum_{2 \leq |\alpha| \leq r-1} \frac{h_0^{2|\alpha|}}{|E_0|} \int_{E_0} |\mathcal{D}^\alpha P(\mathbf{x})|^2 d\mathbf{x}.$$

By the trivial inequality  $P^2 \leq 2((u - P)^2 + u^2)$ , the latter sum is

$$\begin{aligned} & \sum_{2 \leq |\alpha| \leq r-1} \frac{h_0^{2|\alpha|}}{|E_0|} \int_{E_0} |\mathcal{D}^\alpha P(\mathbf{x})|^2 d\mathbf{x} \\ & \leq 2 \sum_{2 \leq |\alpha| \leq r-1} \frac{h_0^{2|\alpha|}}{|E_0|} \int_{E_0} (|\mathcal{D}^\alpha (u(\mathbf{x}) - P(\mathbf{x}))|^2 + |\mathcal{D}^\alpha u(\mathbf{x})|^2) d\mathbf{x} \\ & \leq C \sum_{2 \leq |\alpha| \leq r-1} h_0^{2|\alpha|} (h^{2(r-|\alpha|)} + 1) \\ & \leq Ch^4, \end{aligned}$$

wherein we used (2.6). Moreover,

$$\begin{aligned} & \frac{h_0^2}{|E_0|} \int_{E_0} |\nabla P(\mathbf{x})|^2 d\mathbf{x} \\ &= \frac{h_0^2}{|E_0|} \left( \int_{E_0} |\nabla u(\mathbf{x})|^2 d\mathbf{x} + \int_{E_0} (\nabla P(\mathbf{x}) + \nabla u(\mathbf{x})) \cdot (\nabla P(\mathbf{x}) - \nabla u(\mathbf{x})) d\mathbf{x} \right) \\ &= Dh_0^2 + \frac{h_0^2}{|E_0|} \int_{E_0} (\nabla P(\mathbf{x}) - \nabla u(\mathbf{x}) + 2\nabla u(\mathbf{x})) \cdot (\nabla P(\mathbf{x}) - \nabla u(\mathbf{x})) d\mathbf{x}. \end{aligned}$$

Finally, the last term above, using (2.6) again, is bounded by

$$\begin{aligned} & \frac{h_0^2}{|E_0|} \int_{E_0} (|\nabla P(\mathbf{x}) - \nabla u(\mathbf{x})| + 2|\nabla u(\mathbf{x})|) |\nabla P(\mathbf{x}) - \nabla u(\mathbf{x})| d\mathbf{x} \\ & \leq Ch_0^2 [(h_0^{r-1} + 1) h_0^{r-1}] \leq Ch_0^{r+1} \leq Ch^3, \end{aligned}$$

and the result follows.  $\square$

### 3.1.2 Efficient implementation of $\sigma_{JS}$

Unfortunately, the evaluation of  $\sigma_{JS}$  is a major computational cost in dimensions greater than one over general stencils. The derivatives of  $P$  must be evaluated at specific points and then integrated using numerical integration.

Perhaps Horner's method is the most efficient algorithm to compute the first derivative of a polynomial  $p(t) = \sum_{i=0}^n a_i t^i$  in one dimension. Generalization to higher order derivatives is straightforward, and given in Algorithm 3.2 (see also [5]). Recursion over the variables can be used to handle polynomials in higher dimensions.

#### Algorithm 3.2 Horner's Method for All Derivatives

**Input**  $x \in \mathbb{R}$ , polynomial  $p(t) = \sum_{i=0}^n a_i t^i$   
 $P(d) = 0$  for  $d = 0, 1, \dots, n$   
**for**  $i = n, n-1, \dots, 0$  **do**  
    **for**  $d = n-i, n-i-1, \dots, 1$  **do**  
         $P(d) = P(d)x + dP(d-1)$   
    **end for**  
     $P(0) = P(0)x + a_i$   
**end for**  
**Output**  $P(d) = p^{(d)}(x)$  for  $d = 0, 1, \dots, n$

*Proof* To see that indeed  $P(d) = p^{(d)}(x)$  for  $d = 0, 1, \dots, n$  in Algorithm 3.2, one can proceed by induction on  $n$ . The case  $n = 0$  is trivial, giving only  $P(0) = a_0 = p(x)$ .

If the result holds for  $n-1$ , one can show that it holds for  $n \geq 1$  using the factorization  $p(t) = q(t)t + a_0$ , where  $q(t) = \sum_{i=0}^{n-1} a_{i+1} t^i$ . The algorithm applied to  $q(t)$  results in the derivatives  $P(d) = q^{(d)}(x)$ . For clarity, we rewrite the algorithm for  $q \in \mathbb{P}_{n-1}$ , with a change in index  $j = i+1$  as follows.

$P(d) = 0$  for  $d = 0, 1, \dots, n-1$   
**for**  $j = n, \dots, 1$  **do**



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for  $d = n - j, n - j - 1, \dots, 1$  do
     $P(d) = P(d)x + dP(d-1)$ 
end for
 $P(0) = P(0)x + a_j$ 
end for

```

To treat  $p(x)$ , the algorithm continues by setting  $P(n) = 0$  and taking  $j$  down to 0. The extra loop for  $j = 0$  results in  $P(0) = p(x)$  by the usual Horner's algorithm and

$$P(d) = P(d)x + dP(d-1) = q^{(d)}(x)x + dq^{(d-1)}(x) = p^{(d)}(x),$$

for  $d = 1, \dots, n$ . □

### 3.2 A smoothness indicator $\sigma_P$ based on the polynomial coefficients

We now present an approximation to  $\sigma_{JS}$  on the stencil  $S$  for target element  $E_0 \in S$  that is significantly more efficient to compute. With the expansion of  $P$  given in (2.3), note that (3.2) is

$$\sigma_{JS} = \sum_{1 \leq |\beta| \leq r-1} \frac{h_0^{2|\beta|}}{|E_0|} \int_{E_0} \left( \sum_{|\alpha| < r} c_\alpha \mathcal{D}^\beta \left( \frac{\mathbf{x} - \mathbf{x}_S}{h_S} \right)^\alpha \right)^2 d\mathbf{x}. \quad (3.5)$$

Our first approximation is given by dropping the cross terms to define

$$\begin{aligned} \tilde{\sigma}_P &= \sum_{0 < |\beta| < r} \frac{h_0^{2|\beta|}}{|E_0|} \int_{E_0} \sum_{|\alpha| < r} \left( c_\alpha \mathcal{D}^\beta \left( \frac{\mathbf{x} - \mathbf{x}_S}{h_S} \right)^\alpha \right)^2 d\mathbf{x} \\ &= \sum_{|\alpha| < r} \sum_{0 < \beta \leq \alpha} \left( \frac{h_0}{h_S} \right)^{2|\beta|} c_\alpha^2 \left( \frac{\alpha!}{(\alpha - \beta)!} \right)^2 \frac{1}{|E_0|} \int_{E_0} \left( \frac{\mathbf{x} - \mathbf{x}_S}{h_S} \right)^{2(\alpha - \beta)} d\mathbf{x}. \end{aligned} \quad (3.6)$$

Our second approximation involves the integral over  $E_0$ . We take the point of view that the polynomial is smooth or not on its entire stencil, and so replace  $E_0$  by an idealized target quasi-element  $[-h_S/2, h_S/2]^d + \mathbf{x}_S$ . This leads to the approximation

$$\begin{aligned} \frac{1}{|E_0|} \int_{E_0} \left( \frac{\mathbf{x} - \mathbf{x}_S}{h_S} \right)^{2(\alpha - \beta)} d\mathbf{x} &\approx \int_{[-1/2, 1/2]^d} \mathbf{y}^{2(\alpha - \beta)} d\mathbf{y} \\ &= \prod_{n=1}^d \frac{1}{(2(\alpha_n - \beta_n) + 1) 4^{\alpha_n - \beta_n}}. \end{aligned} \quad (3.7)$$

We define our smoothness indicator based on the polynomial coefficients as

$$\sigma_P = \sum_{|\alpha| < r} \sum_{0 < \beta \leq \alpha} \left( \frac{h_0}{h_S} \right)^{2|\beta|} c_\alpha^2 \left( \frac{\alpha!}{(\alpha - \beta)!} \right)^2 \prod_{n=1}^d \frac{1}{(2(\alpha_n - \beta_n) + 1) 4^{\alpha_n - \beta_n}}. \quad (3.8)$$

The requirement  $0 < \beta$  merely says that the multi-index  $\beta \neq 0$ , so in fact

$$\sigma_P = \sum_{|\alpha| < r} \left[ \sum_{\beta \leq \alpha} \left( \frac{h_0}{h_S} \right)^{2|\beta|} \left( \frac{\alpha!}{(\alpha - \beta)!} \right)^2 \prod_{n=1}^d \frac{1}{(2(\alpha_n - \beta_n) + 1)4^{\alpha_n - \beta_n}} - \prod_{n=1}^d \frac{1}{(2\alpha_n + 1)4^{\alpha_n}} \right] c_\alpha^2. \quad (3.9)$$

For  $i = 0, 1, \dots, r-1$ , let

$$\xi_i = \sum_{k=0}^i \left( \frac{h_0}{h_S} \right)^{2k} \left( \frac{i!}{(i-k)!} \right)^2 \frac{1}{(2(i-k) + 1)4^{i-k}}. \quad (3.10)$$

Then

$$\sigma_P = \sum_{|\alpha| < r} \left( \prod_{n=1}^d \xi_{\alpha_n} - \prod_{n=1}^d \frac{1}{(2\alpha_n + 1)4^{\alpha_n}} \right) c_\alpha^2. \quad (3.11)$$

### 3.2.1 An estimate of $\sigma_P$ in the smooth case

**Lemma 3.3** Suppose  $u$  is approximated on the stencil  $S^{(r)}$  by the polynomial  $P \in \mathbb{P}_{r-1}$ . If  $r \geq 2$ , then as  $h \rightarrow 0^+$ ,

$$\sigma_P = Dh_0h_S + \mathcal{O}(h^4), \quad \text{if } u \text{ is smooth on the stencil,} \quad (3.12)$$

where

$$D = |\nabla u(\mathbf{x}_S)|^2 \geq 0. \quad (3.13)$$

*Proof* When evaluated at the base point  $\mathbf{x} = \mathbf{x}_S$ , we have that

$$c_\alpha = \frac{h_S^{|\alpha|}}{\alpha!} \mathcal{D}^\alpha p(\mathbf{x}_S). \quad (3.14)$$

From (3.10), note that  $\xi_0 = 1$  and  $\xi_1 = 1/12 + h_0/h_S$ . Substitute these results in (3.11) to see that

$$\begin{aligned} \sigma_P &= \sum_{|\alpha| < r} \left( \prod_{n=1}^d \xi_{\alpha_n} - \prod_{n=1}^d \frac{1}{(2\alpha_n + 1)4^{\alpha_n}} \right) \left( \frac{h_S^{|\alpha|}}{\alpha!} \right)^2 (\mathcal{D}^\alpha p(\mathbf{x}_S))^2 \\ &= 0 + \sum_{|\alpha|=1} h_0h_S (\mathcal{D}^\alpha p(\mathbf{x}_S))^2 + \sum_{1 < |\alpha| < r} \mathcal{O}(h^{2|\alpha|}), \end{aligned}$$

which is our desired conclusion.  $\square$

## 4 ML-WENO Reconstructions

Our proposed selection of accurate stencil polynomials leads to a possible reduction in the polynomial degree. This would be unacceptable for standard WENO and WENO-AO reconstructions, which both require exactly two levels of approximation. We now develop a new multilevel WENO (ML-WENO) reconstruction which will be able to combine stencil polynomials of any order.

#### 4.1 The reconstruction

Given a target element  $E_0 \in \mathcal{T}_h$ , select a nonempty collection of stencils  $S_\ell^{(r_\ell)} \ni E_0$ ,  $\ell = 0, 1, \dots, L$ . Each stencil has its polynomial  $P_\ell \in \mathbb{P}_{r_\ell-1}$  approximating  $u$ , as in (2.6). Let the linear weights  $\omega_\ell > 0$  be chosen, one for each stencil. The reconstruction of  $u$  for the collection of stencils is then

$$R(\mathbf{x}) = \sum_{\ell} \tilde{\omega}_\ell P_\ell(\mathbf{x}), \quad (4.1)$$

where  $\tilde{\omega}_\ell$  is the  $\ell$ -th nonlinear weight, which we now define.

We begin by scaling each linear weight. For some choice of parameters  $\varepsilon > 0$ ,  $s \geq 1/2$ , and  $\eta_\ell \geq 0$ , the scaled weight is

$$\hat{\omega}_\ell = \frac{\omega_\ell}{(\sigma_\ell + \varepsilon h_0^2)^{sr_\ell}} \left( \frac{\varepsilon h_0}{\sigma_\ell + \varepsilon h_0} \right)^{\eta_\ell}. \quad (4.2)$$

We suggest taking  $\varepsilon = 0.01$ ,  $s = 1$  or  $2$ , and  $\eta_\ell = 0$ . We then define the nonlinear weight as

$$\tilde{\omega}_\ell = \frac{\hat{\omega}_\ell}{\sum_k \hat{\omega}_k}. \quad (4.3)$$

Provided that (3.1) holds, note that

$$\frac{\varepsilon h_0}{\sigma_\ell + \varepsilon h_0} = \begin{cases} 1 + \mathcal{O}(h) & \text{if } u \text{ is smooth on the stencil } S_\ell^{(r_\ell)}, \\ \Theta(h) & \text{if } u \text{ has a jump discontinuity on the stencil.} \end{cases} \quad (4.4)$$

Therefore, the parameter  $\eta_\ell \geq 0$  can be chosen to increase the bias away from stencils containing solution discontinuities (cf. Lemmas 4.1–4.2 below).

#### 4.2 The ML-WENO Reconstruction error

In this section we consider the accuracy of the ML-WENO reconstruction under the assumption that either  $u$  is smooth or that  $u$  has a jump discontinuity on some, but not all, of the stencils.

To analyze the error in the reconstruction (4.1), we need to consider the many possibilities of where  $u$  might be discontinuous. We find it convenient to divide the stencil indices into disjoint subsets as follows. Let

$$\text{Smooth}(r) = \{\ell : r_\ell = r \text{ and } u \text{ is smooth on } S_\ell^{(r_\ell)}\}, \quad (4.5)$$

$$\text{Jump}(r) = \{\ell : r_\ell = r \text{ and } u \text{ has a jump discontinuity on } S_\ell^{(r_\ell)}\}. \quad (4.6)$$

We tacitly assume that there are no other possibilities, so

$$\bigcup_r (\text{Smooth}(r) \cup \text{Jump}(r)) = \{0, 1, \dots, L\}.$$

We also tacitly assume that  $\bigcup_r \text{Smooth}(r)$  is not empty.

We shall prove that the reconstruction is accurate on the target element  $E_0$  to order

$$r_{\max} = \max \left\{ r : \text{Smooth}(r) \neq \emptyset \right\} \geq 1. \quad (4.7)$$

This is the index of the best (maximal  $r$ ) stencil approximation that avoids the discontinuities in  $u$ . We further define the sets of indices

$$\text{SM} = \text{Smooth}(r_{\max}), \quad \text{SL} = \bigcup_{r < r_{\max}} \text{Smooth}(r), \quad \text{Jmp} = \bigcup_r \text{Jump}(r). \quad (4.8)$$

We begin by estimating the scale factor in (4.2).

**Lemma 4.1** *If (3.1) holds, then*

$$\hat{\omega}_\ell = \begin{cases} \frac{\omega_\ell}{(D + \varepsilon)^{sr_\ell}} h_0^{-2sr_\ell} (1 + \mathcal{O}(h)) & \text{if } \ell \in \text{SM} \cup \text{SL}, \\ \Theta(h^{\eta_\ell}) & \text{if } \ell \in \text{Jmp}. \end{cases} \quad (4.9)$$

We interject that in the proofs of this section, we will make use of well-known results from asymptotic theory (see, e.g., [10]). Specifically, when  $r \geq t$ ,  $a > 0$ , and  $b > 0$ ,

$$\frac{a}{bh^t + \mathcal{O}(h^r)} = \frac{ah^{-t}}{b} (1 + \mathcal{O}(h^{r-t})), \quad (4.10)$$

$$\frac{a}{bh^{-r} + \mathcal{O}(h^{-t})} = \frac{ah^r}{b} (1 + \mathcal{O}(h^{r-t})). \quad (4.11)$$

*Proof* If  $\ell \in \text{SM} \cup \text{SL}$ , then using (3.1) and (4.10), we see that

$$\begin{aligned} \hat{\omega}_\ell &= \frac{\omega_\ell}{(\sigma_\ell + \varepsilon h_0^2)^{sr_\ell}} \left( \frac{\varepsilon h_0}{\sigma_\ell + \varepsilon h_0} \right)^{\eta_\ell} \\ &= \frac{\omega_\ell}{((D + \varepsilon)h_0^2 + \mathcal{O}(h^3))^{sr_\ell}} \left( \frac{\varepsilon h_0}{\varepsilon h_0 + \mathcal{O}(h^2)} \right)^{\eta_\ell} \\ &= \omega_\ell \left( \frac{h_0^{-2}}{D + \varepsilon} (1 + \mathcal{O}(h)) \right)^{sr_\ell} (1 + \mathcal{O}(h))^{\eta_\ell} \\ &= \frac{\omega_\ell}{(D + \varepsilon)^{sr_\ell}} h_0^{-2sr_\ell} (1 + \mathcal{O}(h)) \end{aligned}$$

as claimed. If  $\ell \in \text{Jmp}$ , then

$$\hat{\omega}_\ell = \frac{\omega_\ell}{(\sigma_\ell + \varepsilon h_0^2)^{sr_\ell}} \left( \frac{\varepsilon h_0}{\sigma_\ell + \varepsilon h_0} \right)^{\eta_\ell} = \Theta(h^{\eta_\ell}),$$

and the proof is complete.  $\square$

The result (4.9) is what we need to continue our theoretical development.

**Lemma 4.2** Let  $r_{\max}$  be defined by (4.7). If  $h_0 = \Theta(h)$  and (3.1) and (4.9) hold, then

$$\tilde{\omega}_\ell = \begin{cases} \frac{\omega_\ell}{\sum_{k \in \text{SM}} \omega_k} (1 + \mathcal{O}(h)) & \text{if } \ell \in \text{SM}, \\ \mathcal{O}(h^{2s(r_{\max} - r_\ell)}) & \text{if } \ell \in \text{SL}, \\ \mathcal{O}(h^{2sr_{\max} + \eta_\ell}) & \text{if } \ell \in \text{Jmp}. \end{cases} \quad (4.12)$$

*Proof* The denominator of (4.3) is, using (4.9),

$$\begin{aligned} \sum_\ell \hat{\omega}_\ell &= \sum_{\ell \in \text{SM}} \hat{\omega}_\ell + \sum_{\ell \in \text{SL}} \hat{\omega}_\ell + \sum_{\ell \in \text{Jmp}} \hat{\omega}_\ell \\ &= \sum_{\ell \in \text{SM}} \frac{\omega_\ell}{(D + \varepsilon)^{sr_{\max}}} h_0^{-2sr_{\max}} (1 + \mathcal{O}(h)) + \sum_{\ell \in \text{SL}} \mathcal{O}(h^{-2sr_\ell}) + \sum_{\ell \in \text{Jmp}} \mathcal{O}(h^{\eta_\ell}) \\ &= \sum_{\ell \in \text{SM}} \frac{\omega_\ell}{(D + \varepsilon)^{sr_{\max}}} h_0^{-2sr_{\max}} + \mathcal{O}(h^{1-2sr_{\max}}). \end{aligned}$$

Now (4.11) implies that

$$\begin{aligned} \tilde{\omega}_\ell &= \frac{\hat{\omega}_\ell}{\sum_k \hat{\omega}_k} = \frac{\hat{\omega}_\ell}{\sum_{k \in \text{SM}} \frac{\omega_k}{(D + \varepsilon)^{sr_{\max}}} h_0^{-2sr_{\max}} + \mathcal{O}(h^{1-2sr_{\max}})} \\ &= \frac{\hat{\omega}_\ell}{\sum_{k \in \text{SM}} \frac{\omega_k}{(D + \varepsilon)^{2sr_{\max}}} h_0^{-2sr_{\max}} (1 + \mathcal{O}(h))}. \end{aligned}$$

For any case of  $\ell$ , (4.9) gives the order of  $\hat{\omega}_\ell$ , and the result follows.  $\square$

**Theorem 4.3** Let  $R(\mathbf{x})$  be the reconstruction (4.1). If (2.6) and (4.9) hold, then there is a constant  $C > 0$  such that for any  $\mathbf{x} \in E_0$ ,

$$|u(\mathbf{x}) - R(\mathbf{x})| \leq Ch^{r_{\max}}, \quad (4.13)$$

where  $r_{\max}$  is defined by (4.7).

*Proof* Since the nonlinear weights are positive and sum to 1, we have for  $\mathbf{x} \in E_0$  that the reconstruction error is

$$\begin{aligned} |u(\mathbf{x}) - R(\mathbf{x})| &= \left| \sum_\ell \tilde{\omega}_\ell (u(\mathbf{x}) - P_\ell(\mathbf{x})) \right| \\ &\leq \sum_{\ell \in \text{SM}} \tilde{\omega}_\ell |u(\mathbf{x}) - P_\ell(\mathbf{x})| + \sum_{\ell \in \text{SL}} \tilde{\omega}_\ell |u(\mathbf{x}) - P_\ell(\mathbf{x})| + \sum_{\ell \in \text{Jmp}} \tilde{\omega}_\ell |u(\mathbf{x}) - P_\ell(\mathbf{x})|. \end{aligned} \quad (4.14)$$

Combining this with Lemma 4.2 and the polynomial approximation error (2.6), we obtain

$$\begin{aligned} |u(\mathbf{x}) - R(\mathbf{x})| &\leq \sum_{\ell \in \text{SM}} \Theta(1) \mathcal{O}(h^{r_{\max}}) + \sum_{\ell \in \text{SL}} \mathcal{O}(h^{2s(r_{\max} - r_\ell)}) \mathcal{O}(h^{r_\ell}) \\ &\quad + \sum_{\ell \in \text{Jmp}} \mathcal{O}(h^{2sr_{\max} + \eta_\ell}) \mathcal{O}(1) \\ &= \mathcal{O}(h^{r_{\max}}) + \mathcal{O}(h^{r_{\max} + 2s - 1}) + \mathcal{O}(h^{2sr_{\max} + \eta_\ell}), \end{aligned}$$

which is the desired result.  $\square$

The theoretical optimal choice of  $s$  would be  $1/2$ , but this would require in general the computation of a square root. In practice,  $s = 1, 2, 3, \dots$  is more efficient; moreover, it gives additional bias in  $R$  to the best approximations  $P_\ell$  for  $\ell \in \text{SM}$ .

#### 4.3 Alternative linear weight scalings

As the above theory shows, the key to the linear weight scaling is to obtain (4.9), at least up to the value of the constant. There are many alternate scalings that satisfy this property. One might define

$$\hat{\omega}_\ell = \frac{\omega_\ell}{\sigma_\ell^{sr_\ell} + \varepsilon h_0^{2sr_\ell}} \left( \frac{\varepsilon h_0}{\sigma_\ell + \varepsilon h_0} \right)^{\eta_\ell}, \quad (4.15)$$

which is very similar to the scaling we used.

For some constant  $D_0 > 0$ , one might define

$$\hat{\omega}_\ell = \omega_\ell \left( \frac{\sigma_\ell + 2D_0 h_0}{\sigma_\ell + D_0 h_0^2} \right)^{sr_\ell} \left( \frac{\varepsilon h_0}{\sigma_\ell + \varepsilon h_0} \right)^{\eta_\ell}. \quad (4.16)$$

For this form of the scaling, the theoretical optimal choice of  $s$  is  $s = 1$  (rather than  $s = 1/2$ ), and additional bias in  $R$  to the best approximations  $P_\ell$  for  $\ell \in \text{SM}$  is accomplished by taking a large  $D_0$  (say 10) and positive  $\eta_\ell$ .

#### 4.4 A heuristic explanation of the scaling

In practical application, the reconstruction is used to approximate the viscosity solution of a partial differential equation. As such, it does not have jump discontinuities, but rather steep fronts. We bound the error on the target element in the  $L^2$ -norm as

$$\begin{aligned} \|u - R\|_{L^2} &= \left\| \sum_\ell \tilde{\omega}_\ell (u - P_\ell) \right\|_{L^2} \leq \sum_\ell \tilde{\omega}_\ell \|u - P_\ell\|_{L^2} \\ &\leq \left( \sum_\ell \tilde{\omega}_\ell \|u - P_\ell\|_{L^2}^{2s} \right)^{1/2s} \leq \left( \sum_\ell \tilde{\omega}_\ell (C_A \|u\|_{H^{r_\ell}} h^{r_\ell})^{2s} \right)^{1/2s}, \end{aligned} \quad (4.17)$$

for any  $s \geq 1/2$ , using (2.6).

To proceed, we make the ansatz that  $\sigma_\ell$  measures the smoothness asymptotically according to

$$\|u\|_{H^{r_\ell}} h^{r_\ell} \sim C_B (\sigma_\ell + \varepsilon h_0^2)^{r_\ell/2}. \quad (4.18)$$

Note that if  $u$  is smooth and not constant, both sides are  $\Theta(h^{r_\ell})$ . If  $u$  is not smooth, we might consider both sides to be  $\Theta(1)$ , since we expect  $\mathcal{O}(1)$  approximation of  $u$  by  $P_\ell$ .

With the ansatz,

$$\|u - R\|_{L^2} \sim C_A C_B \left( \sum_\ell \tilde{\omega}_\ell (\sigma_\ell + \varepsilon h_0^2)^{sr_\ell} \right)^{1/2s}. \quad (4.19)$$

We would like to choose  $\tilde{\omega}_\ell$  to be

$$\check{\omega}_\ell = \frac{\omega_\ell h^{2sr_{\max}}}{(C_A C_B)^{2s} (\sigma_\ell + \varepsilon h_0^2)^{sr_\ell}}, \quad (4.20)$$

but we do not know  $r_{\max}$ ,  $C_A$ , and  $C_B$ , and  $\sum_\ell \tilde{\omega}_\ell \neq 1$ . However, we need to normalize, so instead we let

$$\tilde{\omega}_\ell = \frac{\check{\omega}_\ell}{\sum_k \check{\omega}_k} = \frac{\hat{\omega}_\ell}{\sum_k \hat{\omega}_k}, \quad (4.21)$$

where

$$\hat{\omega}_\ell = \frac{\omega_\ell}{(\sigma_\ell + \varepsilon h_0^2)^{sr_\ell}}, \quad (4.22)$$

which is computable since the unknown  $r_{\max}$  and  $C_A C_B$  do not appear.

## 5 Numerical Tests of the ML-WENO Reconstruction

### 6 Application to Conservation Laws

We apply the ML-WENO reconstruction to solve the conservation law

$$u_t(\mathbf{x}, t) + \nabla \cdot f(u; \mathbf{x}, t) = 0, \quad \mathbf{x} \in \Omega, \quad t > 0, \quad (6.1)$$

$$u(\mathbf{x}, 0) = u_0(\mathbf{x}), \quad \mathbf{x} \in \Omega, \quad (6.2)$$

where the vector flux function  $f(u; \mathbf{x}, t)$  and initial condition  $u_0(\mathbf{x})$  are given. Let  $\mathbf{v}$  denote the outer unit normal vector to  $\partial\Omega$ . We can specify an inflow Dirichlet boundary condition

$$u(\mathbf{x}, t) = u_D(\mathbf{x}, t), \quad \mathbf{x} \in \partial\Omega, \quad t > 0, \quad \text{when } f(u) \cdot \mathbf{v} < 0, \quad (6.3)$$

for some specified  $u_D(\mathbf{x}, t)$ , or an inflow flux boundary condition

$$f(u; \mathbf{x}, t) \cdot \mathbf{v} = f_B(\mathbf{x}, t), \quad \mathbf{x} \in \partial\Omega, \quad t > 0, \quad \text{when } f(u) \cdot \mathbf{v} < 0, \quad (6.4)$$

for some specified flux  $f_B(\mathbf{x}, t)$ , or periodic boundary conditions (if  $\Omega$  is suitable).

The semidiscrete finite volume approximation is

$$\bar{u}_{E,t} + \frac{1}{|E|} \int_{\partial E} \hat{f}(u^-, u^+) \cdot \mathbf{v}_E d\sigma(\mathbf{x}) = 0, \quad (6.5)$$

where  $\mathbf{v}_E$  is the outer unit normal vector to  $\partial E$  and  $u^-$  and  $u^+$  are left and right limits of the approximate solution at the interface  $\partial E$ , given by our ML-WENO reconstruction of the element averages  $\bar{u}_E$ . We incorporated a numerical flux  $\hat{f}$ , such as the Lax-Friedrichs one,

$$\hat{f}(u^-, u^+; \mathbf{x}, t) = \frac{1}{2} [(f(u^-; \mathbf{x}, t) + f(u^+; \mathbf{x}, t)) \cdot \mathbf{v}_E - \alpha_{\text{LF}}(u^+ - u^-)], \quad (6.6)$$

where

$$\alpha_{\text{LF}} = \max_u |\partial f / \partial u| \quad (6.7)$$

is the maximum wave speed.

Time is discretized at the time levels  $0 = t^0 < t^1 < t^2 < \dots$  and a Runge-Kutta time integrator is employed. In the sequel, we compute  $\bar{u}_E^n \approx \bar{u}_E(t^n)$  for each  $n > 0$ .

## 7 Summary and Conclusions

### References

1. Arbogast, T., Huang, C.S., Zhao, X.: Accuracy of WENO and adaptive order WENO reconstructions for solving conservation laws. *SIAM J. Numer. Anal.* **56**(3), 1818–1847 (2018). DOI 10.1137/17M1154758
2. Balsara, D.S., Garain, S., Florinski, V., Boscheri, W.: An efficient class of weno schemes with adaptive order for unstructured meshes. *J. Comput. Phys.* **404**, 109062 (2020). DOI 10.1016/j.jcp.2019.109062
3. Balsara, D.S., Garain, S., Shu, C.W.: An efficient class of WENO schemes with adaptive order. *J. Comput. Phys.* **326**, 780–804 (2016)
4. Bramble, J.H., Hilbert, S.R.: Estimation of linear functionals on Sobolev spaces with applications to Fourier transforms and spline interpolation. *SIAM J. Numer. Anal.* **7**, 112–124 (1970)
5. Carnicer, J., Gasca, M.: Evaluation of multivariate polynomials and their derivatives. *Math. Comp.* **54**(189), 231–243 (1990)
6. Dupont, T., Scott, L.R.: Polynomial approximation of functions in Sobolev space. *Math. Comp.* **34**, 441–463 (1980)
7. Harten, A., Osher, S.: Uniformly high-order accurate nonoscillatory schemes I. *SIAM J. Numer. Anal.* **24**(2), 279–309 (1987)
8. Jiang, G.S., Shu, C.W.: Efficient implementation of weighted ENO schemes. *J. Comput. Phys.* **126**, 202–228 (1996)
9. Liu, X.D., Osher, S., Chan, T.: Weighted essentially non-oscillatory schemes. *J. Comput. Phys.* **115**, 200–212 (1994)
10. Olver, F.W.J.: *Introduction to Asymptotics and Special Functions*. Academic Press, New York (1974)
11. Shu, C.W., Osher, S.: Efficient implementation of essentially non-oscillatory shock capturing schemes. *J. Comput. Phys.* **77**, 439–471 (1988)